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PROJECT REPORT

HOMEFINDER

Real Estate Assistance Web Application

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Course: Bachelor of Computer Applications (BCA)

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A structured web portal for finding and managing residential property listings, client requirements, and meeting requests.

CERTIFICATE

This is to certify that the project titled **"HomeFinder - Real Estate Assistance Portal"** is a genuine work carried out by **Sakshi Shashikant Khopade** in partial fulfillment of the requirements for the award of the degree of **Bachelor of Computer Applications (BCA)**.

The project has been completed under the guidance of **Prof. Monali Meghal** and is submitted for evaluation.

Student Signature	Guide Signature	Head of Department

DECLARATION

I hereby declare that this project report titled **"HomeFinder - Real Estate Assistance Portal"** is my original work and has not been submitted previously in any form for the award of any degree, diploma, or other qualification.

Name: Sakshi Shashikant Khopade

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Signature: ____________________

Date: ____________________

ACKNOWLEDGEMENT

I express my sincere gratitude to my project mentor **Prof. Monali Meghal** for the continuous support, valuable suggestions, and guidance throughout the development of this project.

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INDEX

1. Abstract

HomeFinder is a web-based real estate assistance portal designed to help users discover residential properties for rent or purchase in a structured and efficient way. The platform reduces the friction commonly experienced in property search by collecting user requirements in a standardized format, matching them against available listings, and enabling a controlled interest-confirmation flow for serious users.

The application supports three roles: guest visitor, registered client, and administrator. Guests can submit a quick requirement form and view a limited preview of matches. Registered clients can save detailed preferences, view full property details, mark interest in properties, and confirm a meeting through a simulated payment flow. Administrators can manage listings, view clients, track interests, and update meeting status.

2. Introduction

Property search in many cities is fragmented across classified portals, brokers, and informal channels. This often results in repeated conversations, irrelevant suggestions, and poor lead tracking. HomeFinder addresses that problem by collecting requirements in a structured form, filtering listings by budget, area, BHK, and amenities, and giving administrators a single place to manage leads and property data.

The application is designed to be simple enough for academic evaluation while still reflecting real-world system thinking.

3. Problem Statement

Traditional real-estate discovery is time-consuming and repetitive. A user usually has to search multiple websites, contact brokers individually, repeat requirements many times, and follow up manually for site visits.

At the same time, property agencies struggle to track genuine leads, payment confirmations, and meeting schedules in a centralized way. HomeFinder solves these issues by using requirement-based search, session-based authentication, admin-controlled listings, an interest-confirmation flow, and centralized status tracking.

4. Objectives

1. Provide a guest-friendly property search experience without mandatory login.
2. Allow registered clients to save detailed preferences and receive better matches.
3. Enable property discovery by area, transaction type, BHK, budget, and amenities.
4. Support interest confirmation through a simulated payment flow.
5. Give administrators control over property listings, client records, and meeting management.
6. Demonstrate secure and structured use of PHP, MySQL, HTML, CSS, and JavaScript.

5. Scope of the Project

In Scope

- Landing page with hero, statistics, how-it-works, and quick form
- Guest requirement submission and limited preview of matches
- Client registration and login
- Client dashboard with preference management
- Property listing and filtering
- Single property detail page
- Interest confirmation and simulated payment flow
- Payment success confirmation page
- Admin login and registration
- Admin dashboard and CRUD pages

Out of Scope

- Live payment gateway integration
- Mobile application
- Real-time chat
- Maps integration
- Automated email/SMS notifications
- Multi-language support
- Owner self-listing portal

6. Users and Roles

Guest User

Guest users can explore the portal, submit a quick requirement form, and view a limited number of matched properties.

Registered Client

Registered clients can save preferences, access full property details, mark interest, and confirm a meeting through the booking flow.

Admin

Administrators manage the overall system, including listings, users, interest records, payment status, and meeting updates.

7. System Requirements

Hardware Requirements

- Processor: Intel Core i3 or above
- RAM: 4 GB minimum, 8 GB recommended
- Storage: 2 GB free space minimum
- Display: 1366x768 or higher

Software Requirements

- Windows or any Apache-compatible environment
- XAMPP with Apache and MySQL
- PHP 8.x
- MySQL 8.x
- Modern browser such as Chrome, Edge, or Firefox

8. Technology Stack

Front End

- HTML5 for semantic structure
- CSS3 for layout, responsiveness, and component styling
- Vanilla JavaScript for validation, counters, filtering, and AJAX interactions

Back End

- PHP 8.x for routing, authentication, validation, and database operations
- MySQL 8.x for persistent storage
- Apache via XAMPP as the web server

9. Functional Features

- Public landing page with hero banner and animated counters
- Quick requirement submission form
- Featured properties section
- Client registration, login, and dashboard
- Property listing, filtering, and detail view
- Interest toggle, simulated payment, and booking confirmation
- Admin login, dashboard, property CRUD, client view, and meeting management

10. Page-Wise Description

Home Page

The home page introduces the platform, displays key metrics, and provides a quick requirement form for guests.

Registration and Login

Users can register with name, email, phone number, and password. Registered clients authenticate through email and password, after which a session is started.

Client Dashboard

The dashboard displays saved preferences and matching properties. Users can also review their saved interests and meeting states.

Property Listing Page

The properties page provides filters for transaction type, BHK, area, price range, amenities, and sorting options. Guests see a limited preview, while logged-in users can browse more fully.

Property Detail Page

The detail page shows the property gallery, description, amenities, location, and a sticky interest call-to-action.

Payment and Confirmation Pages

The payment flow is simulated for project purposes. After confirmation, the system records an interest entry and payment record, then shows a success page with next steps.

Admin Pages

The admin section includes login, registration, dashboard, property management, client management, interest management, and status update functionality.

11. Architecture Overview

HomeFinder follows a simple three-tier architecture. The presentation layer is rendered in the browser using HTML, CSS, and JavaScript. The application layer is implemented in PHP pages and backend handlers that process requests, validate data, and manage sessions. The data layer uses MySQL to store users, admins, properties, preferences, search requests, interests, and payments.

> **Request Flow:** Browser request -> PHP validation -> PDO prepared statements -> MySQL result -> HTML or JSON response -> JavaScript enhancement if needed.

12. Database Design

users: Registered client accounts

admins: Admin accounts separated from clients

preferences: One record per user for saved search preferences

properties: Central property listings table

property_images: Multiple images per property

search_requests: Guest requirement submissions

interests: Confirmed interest and meeting status records

payments: Simulated payment records linked to interests

13. Entity Relationship Summary

- One user has one preferences record.
- One user can create many search requests.
- One user can create many interests.
- One property can receive many interests.
- One interest has one payment record.
- One property can have many images.
- One admin can create many properties.

14. Security and Validation

- Password hashing using `password_hash()`
- Password verification using `password_verify()`
- Session-based authentication
- Session ID regeneration after login
- Prepared statements through PDO
- CSRF token generation and verification
- Server-side validation for required fields and formats
- Restricted admin registration using a secret key
- Safe internal redirects

15. Front-End Design Approach

The UI uses a clean, modern real-estate style with semantic HTML5 markup, responsive CSS Grid and Flexbox layouts, card-based presentation for properties, sticky sidebars for filters and CTAs, animated counters on the home page, and consistent button, form, and badge styling.

The design prioritizes clarity, readability, and mobile responsiveness.

16. JavaScript Functionality

- Form validation
- Animated counters
- Navigation toggles
- Property filter state handling
- AJAX interest toggling
- Payment form interactions
- Image gallery switching on the property detail page

17. Key Modules

includes/init.php: Initializes sessions, headers, timezone, and shared helpers.

php/db.php: Provides the PDO database connection and configuration constants.

php/helpers.php: Contains utility functions for redirects, sanitization, CSRF, formatting, and property matching.

php/auth.php: Handles client and admin login and registration flows.

php/properties.php: Processes property search and CRUD actions.

php/interests.php: Handles interest toggling and payment confirmation.

18. Testing Summary

1. Guest can submit the quick requirement form.
2. Guest preview results are limited to three properties.
3. Client registration stores hashed passwords and starts a session.
4. Client login redirects to the dashboard.
5. Preferences can be saved and reflected in matched results.
6. Property detail page shows protected address to logged-in users only.
7. Interest toggle respects paid interests and does not remove them.
8. Payment confirmation creates both interest and payment records.
9. Admin can add, edit, and delete properties.
10. Admin can filter interest records and update meeting status.

19. Observations

The project successfully demonstrates structured user requirement collection, filtered property discovery, role-based access control, simulated booking/payment flow, admin-driven operations, and database-backed CRUD functionality.

20. Future Enhancements

- Live payment gateway integration
- Email and SMS notifications
- Map-based search
- Chat between client and agent
- Image upload previews
- Analytics charts for admin
- Pagination UX improvements
- Saved shortlist functionality
- PDF report generation for meeting confirmations

21. Conclusion

HomeFinder is a complete real-estate assistance web application built using core web technologies and a relational database. It addresses a practical search and lead-management problem while remaining suitable for a third-year BCA final project.

The system demonstrates the ability to design a multi-role application, implement secure authentication, manage structured data, and build a responsive interface that is both functional and easy to understand.

22. Appendix

Project Structure

```
/homefinder/\n??? index.php\n??? login.php\n??? register.php\n???
dashboard.php\n??? properties.php\n??? property-detail.php\n??? payment.php\n???
payment-success.php\n??? logout.php\n??? admin/\n??? php/\n??? includes/\n???
assets/\n??? sql/
```

Sample Property Locations

- Kothrud
- Baner
- Wakad
- Aundh
- Viman Nagar
- Hadapsar